

Auxiliary equation

Key Points

To solve a 2nd order differential equation of the form

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

The auxiliary equation is

$$am^2 + bm + c = 0$$

Found by trying $y = e^{mx}$

With discriminant $d = b^2 - 4ac$ determining the types of roots and hence the solutions

Basic Skills

Q1. Write the auxiliary equations for the following

a) $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 20y = 0$

b) $\frac{d^2 y}{dx^2} + 9y = 0$

c) $\frac{d^2 y}{dx^2} = 9y$

Q2. Show that both $y = e^x$ and $y = e^{2x}$ solve the differential equation $y'' - 3y' + 2y = 0$ and hence write a general solution

Q3. By substituting $y = e^{mx}$ into $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} - 6y = 0$ show that the auxiliary equation is $m^2 - 5m - 6 = 0$

Competency Skills

Q4. Prove why the auxiliary equation must be true in order to get solutions to $a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$

Q5. Explain what types of solutions can the auxiliary equation have

Q6. For the differential equation $a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$

a) Given that $b^2 - 4ac > 0$ and so 2 real solutions are found write the general solution

b) Given that $b^2 - 4ac = 0$ and so 1 real (repeated) auxiliary solution is found write the general solution

c) Given that $b^2 - 4ac < 0$ and so 2 complex auxiliary solutions are found write the general solution

Q7. Find the general solution to the differential equation $2 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 2y = 0$

Advanced Skills

Q8. For the differential equation $a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$ when the auxiliary equation gives 2 complex solutions $m = R \pm Ii$

a) Explain why a general solution is $y = Ae^{(R+Ii)x} + Be^{(R-Ii)x}$

b) By using Euler's identity convert the general solution into $y = e^{Rx}(C \cos(Ix) + D \sin(Ix))$ stating the relationships between constants A, B and C, D

Q9. For the differential equation $a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$ when the auxiliary equation gives 1 real solution m

a) Explain why initially only $y = Ae^{mx}$ can only be written to solve the differential equation

b) By considering $y = f(x)e^{mx}$ show that the only function $f(x)$ is linear and hence explain why the general solution is $y = Ae^{mx} + Bxe^{mx}$

Extension

Q10. Find a general solution to the fourth order ODE $\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} - 30y = 0$

Hint: Substitution to make second order

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